

Sovereign Industry Foundation Model - Concept Proposal

Post-training a defence-grade industry expert model on 25 years of multi-corporate engineering data

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The Opportunity

Twenty-five years of design, engineering, supply chain, procurement, programme management and support data - petabytes, dozens of formats, spread across multiple corporates - is the single most valuable untapped training corpus in UK industry. No existing model understands your programmes, your parts, your failure modes, your suppliers. A post-trained model built on this corpus becomes a compounding industry asset: an expert that has read everything, forgotten nothing, and answers in your language.

Cosine builds this with **Sparklr**, our training and post-training platform, on top of our own sovereign UK foundation model.

Why Cosine is different

- **Sovereign by construction.** The only large language model trained from scratch in the UK. No US export-control exposure, no foreign jurisdiction over weights, training data or telemetry. Auditable end to end.
- **Defence-grade provenance.** Every training sample is lineage-tracked to source system, owner, classification and licence. Provenance is queryable - you can ask the model *why it knows something* and trace it to the document.
- **Built for classified environments.** Trains and deploys inside your security boundary - air-gapped or accredited cloud. Data never leaves sovereign infrastructure. Personnel and facilities operate to defence industry standards.
- **Sparklr efficiency.** Sparklr's data curation, deduplication across formats, and post-training pipeline cut effective training compute by an order of magnitude versus naive fine-tuning. Faster iteration means more experiments per pound and a shorter path to a model that is genuinely expert, not generic-with-a-lookup.

Delivery roadmap - value early, capability compounding

Phase	Model	Purpose	Timeline
1. Demonstrator	Fine-tune of Cosine base model on a scoped data slice (1-2 domains)	Prove value in weeks; de-risk data pipelines and governance	8-12 weeks from data access

Phase	Model	Purpose	Timeline
2. Interim	Post-trained model on full cross-corporate corpus	Broad industry competence; first production use cases	6-9 months
3. Industry expert	Large-scale post-trained model, full corpus, continuous refresh	The authoritative industry model; deep domain reasoning	12-18 months
4. Sustained	Scheduled fine-tunes on new data + support services	Model stays current as programmes evolve	Multi-year support agreement

Data and training scale (indicative - to be calibrated in Phase 0)

Petabyte-scale raw corpora typically yield 2-10 trillion usable tokens after extraction, cleaning and cross-format deduplication. Indicative training envelopes:

Target	Tokens	Model scale	Training	Training time	Hardware (training)
Demonstrator	~50-100B	8-13B params	~10 ²¹ FLOPs	Days	8-16— H100-class GPUs
Interim	~1-2T	30-70B params	~10 ²³ FLOPs	4-8 weeks	128-256— H100-class GPUs
Industry expert	~3-6T	70-180B params	~10 ²⁴ -10 ²⁵ FLOPs	3-6 months	512-1024— H100-class GPUs

Sparklr's curriculum and data-efficiency techniques consistently reduce these envelopes; final figures are fixed during Phase 0 scoping.

Deployment compute

Tier	Use case	Hardware per instance
Edge / programme site	Single-team assistant, intermittent use	1-2— L40S/H100-class GPUs (quantised)

Tier	Use case	Hardware per instance
Enterprise	Concurrent users across a programme	4-8— H100-class GPUs
Sovereign hub	Multi-corporate, high-assurance serving	Redundant GPU cluster inside accredited boundary, air-gap option

All tiers run entirely within customer-controlled infrastructure. No external API calls.

IP and data governance between contributing corporates

This is the hardest part, and we treat it as a first-class engineering problem, not a legal afterthought.

- **Federated ingestion with attribution.** Each contributor's data is ingested, cleaned and tokenised within its own boundary or accredited enclave. Provenance metadata is bound at token level.
- **Contractual firewalls.** A tripartite framework agreement defines: no contributor can extract another's data from the model; model outputs are filtered against source-attribution rules; competitive-overlap domains can be ring-fenced into partitioned training or excluded.
- **Sovereign escrow.** Weights and corpus held under UK jurisdiction, with pre-agreed exit and dissolution terms - contributors know exactly what happens to their IP if the consortium changes.
- **Asymmetric benefit, symmetric protection.** Each corporate gets a model tuned to its own estate; shared learning accrues in generalised capability, never in disclosable specifics. Technical controls (output attribution, canary tests, membership-inference auditing) verify this continuously.
- **Governance board.** Joint customer board owns accept/reject authority over training runs, releases and use-case expansion.

Commercial structure (indicative)

1. **Phase 0 scoping** (fixed price): data audit, token yield analysis, governance framework, calibrated compute plan.
2. **Model builds** per roadmap phases, milestone-priced.

3. **Compute:** customer-owned, Cosine-managed, or hybrid - we specify, procure and operate as agreed.
4. **Support services:** multi-year agreement covering scheduled fine-tunes on new data, evaluation and red-teaming, platform updates, and named engineering support.

Next step

A two-week Phase 0 workshop: we audit a sample of the corpus, calibrate the numbers in this document against your actual data, and return a costed, scheduled plan for the demonstrator. Contact us to mobilise.